26

Point-of-Care Ultrasound

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KEY POINTS

- Point-of-care ultrasonography (POCUS) constitutes a focused ultrasound study performed by clinicians as part of a physical examination. The effectiveness of POCUS hinges on the proper clinical integration of findings and operator proficiency.
- Increased renal cortical echogenicity is a nonspecific finding, observed in both acute kidney injury and chronic kidney disease. The probability of chronic kidney disease is increased when coupled with reduced cortical thickness and renal length.
- Hydronephrosis can be differentiated from prominent vasculature or vascular malformations by using color Doppler imaging, as the former does not exhibit color flow.
- Left pleural effusion, commonly seen on parasternal cardiac view, may be confused with pericardial effusion. The descending thoracic aorta helps differentiate them: Pericardial effusion is anterior, while pleural effusion extends posterior to the aorta.
- Right ventricular and inferior vena cava dilation can occur in both volume overload and pressure overload (e.g., pulmonary embolism). Interpretation within the clinical context alongside other laboratory and imaging findings is vital.
- B-lines observed on lung ultrasound are not exclusive to pulmonary edema. They may also appear in conditions like pulmonary fibrosis, pneumonia, alveolar hemorrhage, and contusion.
- Ultrasound of the internal jugular vein provides an alternative to the inferior vena cava for estimating right atrial pressure, particularly in situations where the inferior vena cava is inaccessible or deemed unreliable (e.g., cirrhosis).
- The Doppler stigmata of systemic venous congestion do not differentiate between volume and pressure overload, much like inferior vena cava ultrasound. Venous congestion reflects the interplay between right atrial pressure, intravascular volume and venous compliance.

Point-of-care ultrasonography (POCUS) is a focused ultrasound examination performed by the clinician at the bedside to answer specific questions that aid in patient management.[1] Ultrasonography itself has been a trusted imaging tool for several decades. With ongoing technologic progress and the miniaturization of ultrasound devices, clinician-performed ultrasound has gained significant momentum in the recent past, emerging as a fifth pillar of physical examination along with the traditional four pillars, namely inspection, palpation, percussion, and auscultation.[2] POCUS is essentially a bedside diagnostic tool, like the stethoscope, used to make a diagnosis, fine-tune the list of potential diagnoses, or monitor response to treatment. In nephrology practice, examples of focused clinical questions that can be answered by POCUS include, “Does this patient with acute kidney injury (AKI) have hydronephrosis?”, “Does this patient with shortness of breath have pulmonary edema?”, “Is the uremic pericardial effusion improving with intensive dialysis?” etc.[3] Studies have demonstrated that POCUS enhances diagnostic precision, reduces the time to diagnosis, positively impacts physician-patient interactions, and lessens the financial burden on the health care system, and the findings hold prognostic significance.[4,5] Nonetheless, POCUS is not a substitute for consultative ultrasound imaging and users must exercise clinical judgment to determine when a focused examination is sufficient and when a more detailed assessment (e.g., a comprehensive echocardiogram) may be required for making accurate and well-informed clinical decisions. In this chapter, we provide an overview of the key diagnostic POCUS applications relevant to nephrology.

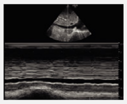
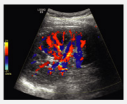
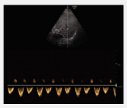
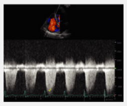
PRINCIPLES AND TERMINOLOGY

Fundamentally, medical ultrasonography involves emission of high-frequency sound waves using piezoelectric crystals within the transducer (probe) and conversion of the returning echoes into electrical signals, which are then used to create images depicting the scanned structures. Some newer handheld devices employ capacitive-micromachined ultrasonic transducers instead of the traditional piezoelectric technology. Nevertheless, the underlying wave properties and principles of image interpretation remain consistent.[6] Table 26.1 summarizes commonly used ultrasound modes for POCUS applications.

With respect to image orientation, structures closer to the transducer are displayed at the top of the image, while structures farther from the transducer appear at the bottom. The orientation marker on the ultrasound screen (screen indicator) corresponds to the orientation marker on the transducer (probe marker). For instance, as the long axis view of the kidney is typically acquired with the probe marker oriented toward the patient's head, the upper pole of the kidney is positioned toward the screen indicator and the lower pole faces the opposite side.

In grayscale imaging, the displayed structures are described on the basis of their echogenicity, which refers to the degree of reflection of sound waves in relation to surrounding structures. Anechoic structures, which transmit all sound waves without reflection, appear black. Examples include fluid-filled structures such as blood vessels, urinary bladder, and cysts. Hypoechoic structures reflect fewer sound waves compared with surrounding structures, as seen with the renal cortex compared with the liver. Isoechoic structures reflect sound waves similarly to surrounding structures. For example, an angiomyolipoma is isoechoic to the renal sinus fat. Hyperechoic structures reflect most sound waves and appear bright or white. Fibrous structures like the diaphragm or pericardium are hyperechoic. Some hyperechoic structures, such as bones and stones cast shadows due to their near-complete obstruction of sound

Table 26.1 Commonly Used Ultrasound Modes and Imaging Techniques

Brightness (B)-mode 	Also referred to as the 2D mode, this represents the standard grayscale imaging mode. Within this mode, the echogenicity, or *brightness*, of the imaged structures is determined by the intensity of the reflected signals.
Motion (M)-mode 	This is an imaging mode used to evaluate and quantify the movement of structures. Within this mode, a unidimensional beam is directed along the cursor's path overlying a 2D image, and the motion of all structures along that line is recorded over time. This facilitates the measurement of cavity dimensions or the movement of structures (e.g., respiratory variation of the inferior vena cava diameter and cardiac valve motion).
Color Flow Doppler 	Color Flow Doppler enables the visualization of blood flow and its directional information. It functions based on the principle of frequency change between the reflected ultrasound waves and the emitted waves. Typically, red on color Doppler signifies flow toward the transducer, while blue indicates flow away from the transducer.
Power Doppler 	Power Doppler is also used to detect blood flow but unlike color Doppler, it does not provide directional information (no red-blue indication). Instead, it conveys information through the hue and brightness of the signal, which represent the *power* or intensity of the returning echoes. It is more sensitive than color Doppler for detecting low flows (e.g., small vessels in the needle pathway while performing a kidney biopsy).
Pulsed Wave Doppler 	Pulsed wave Doppler operates based on the frequency shift like color Doppler but offers a graphical depiction of blood velocity over time at a specific location. There is a limit to the velocity that can be accurately measured as it entails emitting sound waves in discrete *pulses* and the transducer pauses until the returning echo is received. This mode is used to evaluate renal artery stenosis, estimate cardiac output, etc.
Continuous Wave Doppler 	Continuous wave Doppler records flow velocities along the entire Doppler cursor, rendering it unsuitable for pinpointing flow at a specific location. However, because sound waves are emitted and received *continuously* through distinct transducer elements, there is no limitation on velocity, in contrast to pulsed wave Doppler. This mode is used to measure transvalvular flow gradients in echocardiography.

wave transmission, often obscuring underlying structures. Also, air scatters ultrasound waves, resulting in a bright signal with a hazy shadowing effect, which can obstruct the visualization of structures underneath it. While the detailed description of ultrasound physics and artifact generation is beyond the scope of this chapter, it is important to emphasize that the optimal definition of structures is attained in grayscale ultrasound when the ultrasound beam intersects at a perpendicular angle. Conversely, the beam must be parallel to the flow in Doppler imaging for precise velocity measurements.

POCUS OF THE KIDNEY AND URINARY BLADDER

On ultrasound images, the kidney is usually well defined due to the surrounding hyperechoic capsule and appears as a bean-shaped structure (Fig. 26.1). The renal parenchyma consists of two distinct regions: the outer cortex and inner medulla, organized into pyramids. The renal cortex is either isoechoic or more commonly hypoechoic when compared with the normal liver or spleen. On the other hand, the medullary pyramids are hypoechoic or anechoic in comparison with the cortex. The hyperechoic sinus fat occupies a significant portion of the inner kidney space. The collecting system is not easily seen unless it becomes dilated, causing it to push apart the surrounding fat. Medullary pyramids are not well visualized in adults, especially when using low-resolution handheld ultrasound devices. As such, we should not rely heavily on the loss of corticomedullary differentiation when attempting to distinguish between AKI and chronic kidney disease (CKD).

The primary purpose of kidney POCUS is to exclude hydronephrosis. Nevertheless, it is crucial to follow a systematic approach when interpreting the images to ensure that significant abnormalities are not overlooked. A mnemonic format checklist, **SECONDS**, has been proposed to facilitate this methodic interpretation.[7] S stands for size, encompassing renal length and cortical or parenchymal thickness. In adults, the kidney length usually ranges from 10 to 12 cm. Cortical thickness is measured from the base of the medullary pyramid to the outer margin of the kidney. It is usually 0.7 to 1 cm but can vary within a kidney, with thicker measurements observed at the poles. In cases where the medullary pyramids are not visible, parenchymal thickness should be measured. This measurement is taken from the outer margin of the kidney to the sinus fat and typically falls within the range of 1.5 to 2.0 cm.[8] Size of the kidney reduces in CKD except in hyperfiltration states, such as diabetes

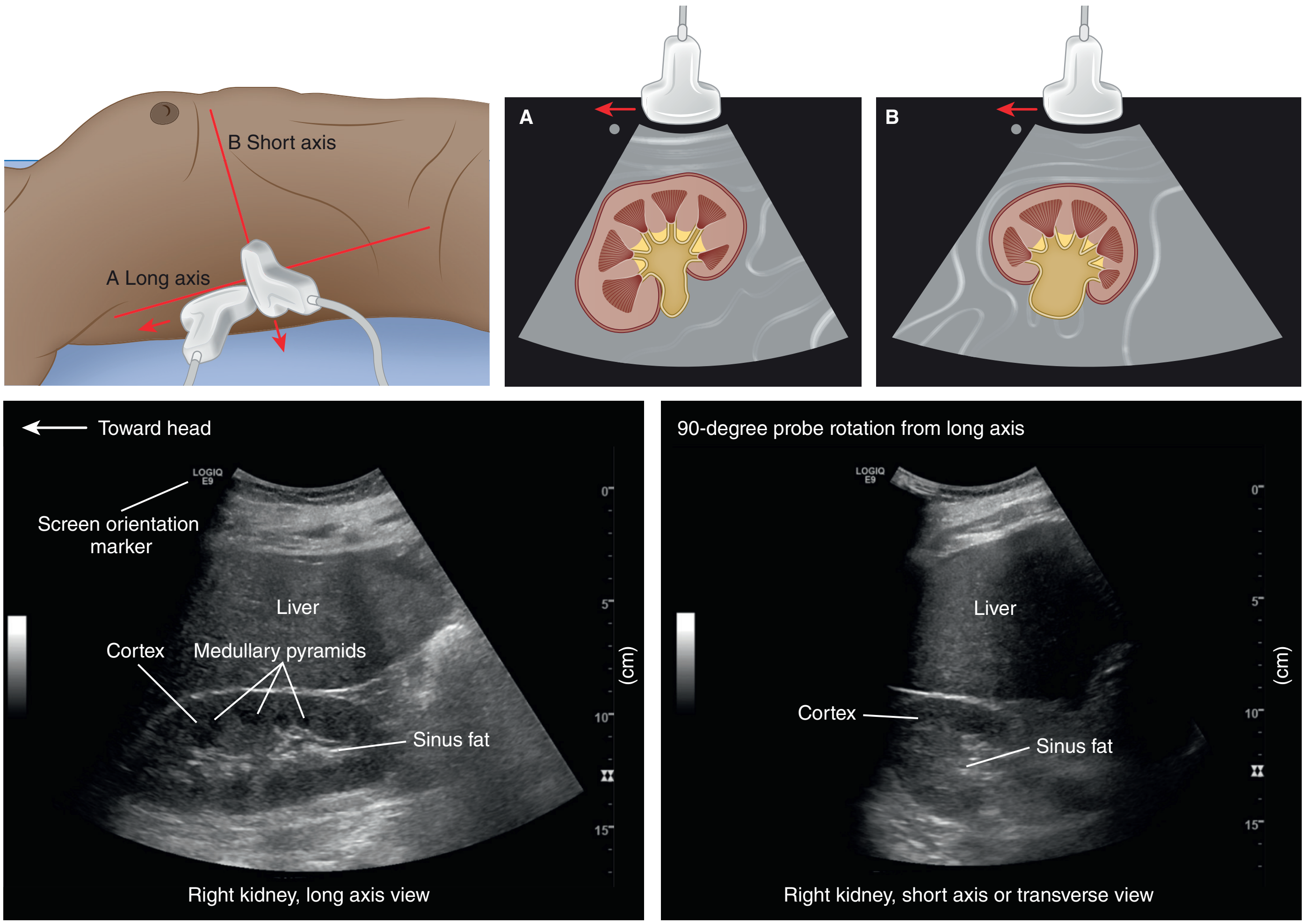


Fig. 26.1 Upper panel: Transducer positions for obtaining long and short-axis views of the kidney, with *arrows* indicating the direction of the transducer orientation marker, along with the corresponding images. Lower panel: Sonographic images of a normal kidney.

mellitus or infiltrative diseases like amyloidosis. E stands for relative echogenicity, indicating how bright the renal cortex appears compared with the adjacent liver or spleen. Elevated cortical echogenicity is a nonspecific finding and can be observed in both AKI (glomerulonephritis, acute tubular necrosis, acute interstitial nephritis) and CKD. However, in general, the likelihood of treatable pathology in small kidneys (<9 cm) with increased cortical echogenicity is low.[9] Fig. 26.2A depicts a kidney with increased cortical echogenicity while maintaining its size, suggesting a higher likelihood of AKI when the baseline kidney function is not available. Renal biopsy in this case confirmed extensive tubular injury. Fig. 26.2B demonstrates a small kidney with reduced length and parenchymal thickness, obtained from a patient with CKD stage 5.

C in the SECONDS checklist indicates collecting system (i.e., presence or absence of hydronephrosis). Hydronephrosis appears as a branching, anechoic area extending into the kidney, usually due to a distal obstruction such as a ureteropelvic stone or a bladder pathology. As the severity of hydronephrosis increases, more urine accumulates within the kidney, exerting pressure on the parenchyma. For POCUS purposes, hydronephrosis is graded as mild, moderate, and severe. In mild cases, there is dilation of the renal pelvis and calyces, but the overall pelvicalyceal pattern remains intact (Fig. 26.3A). In moderate hydronephrosis, the pelvis enlarges further and the calyces become rounded and fused giving a cauliflower-like appearance (Fig. 26.3B). In severe cases, the renal pelvis and calyces become further enlarged and parenchymal thinning is often noted (Fig. 26.3C).[10] Since this grading system is qualitative, it is acceptable to state findings as "mild to moderate" or "moderate to severe." Certain conditions can mimic hydronephrosis, and it is crucial for POCUS users to be aware of them. Prominent renal vessels or vascular malformations may appear as anechoic branching structures in the renal pelvis area, but color Doppler distinguishes them from collecting system dilation (Fig. 26.4A and B). The key is to evaluate any anechoic structure using color Doppler before calling it a dilated collecting

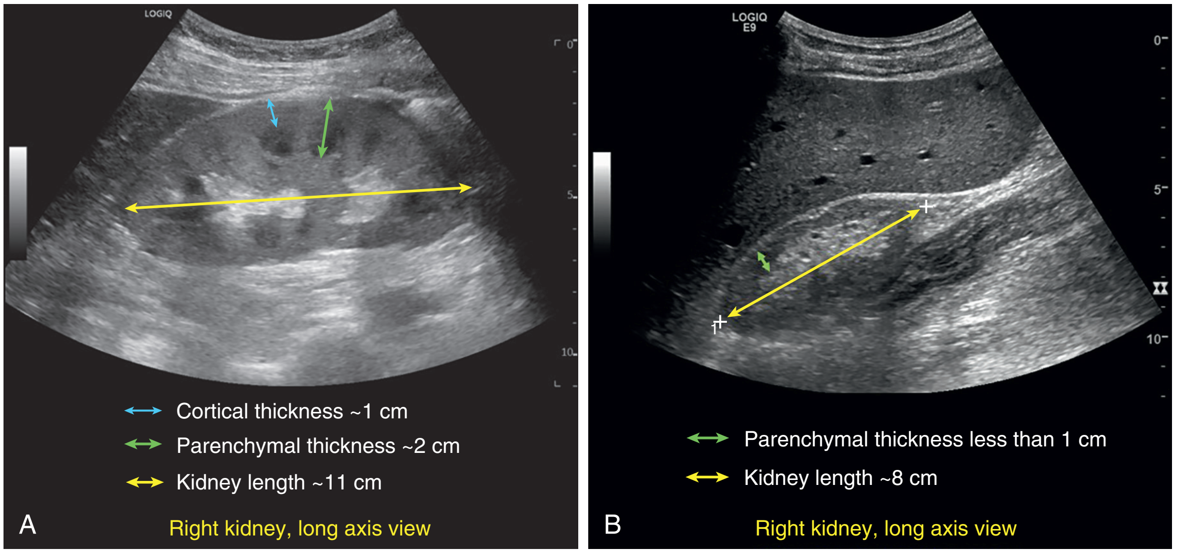


Fig. 26.2 Kidney ultrasound demonstrating (A) increased cortical echogenicity with preserved size and (B) short, thin kidney with increased cortical echogenicity.

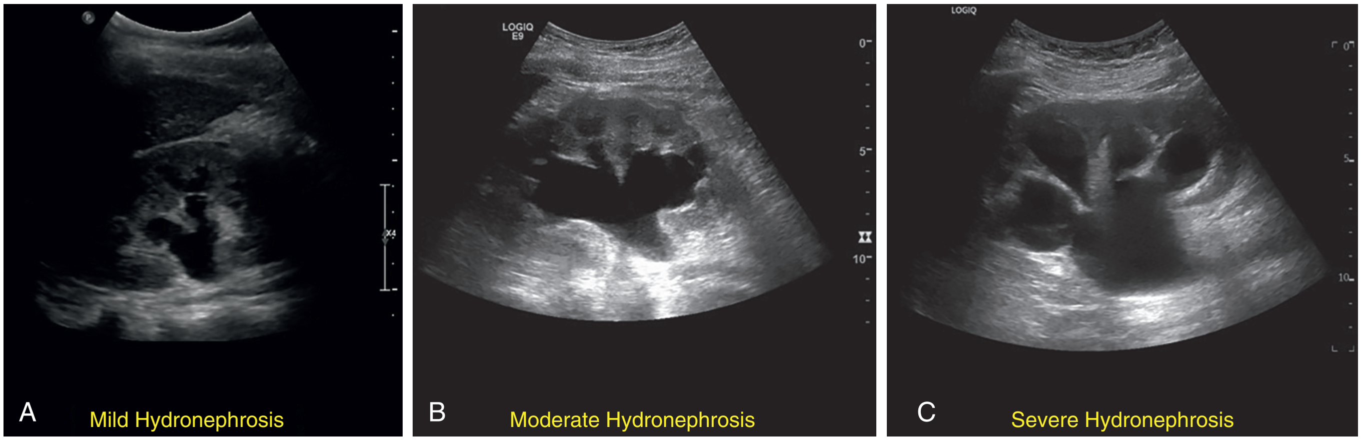


Fig. 26.3 Qualitative grading of hydronephrosis: (A) mild, (B) moderate, and (C) severe.

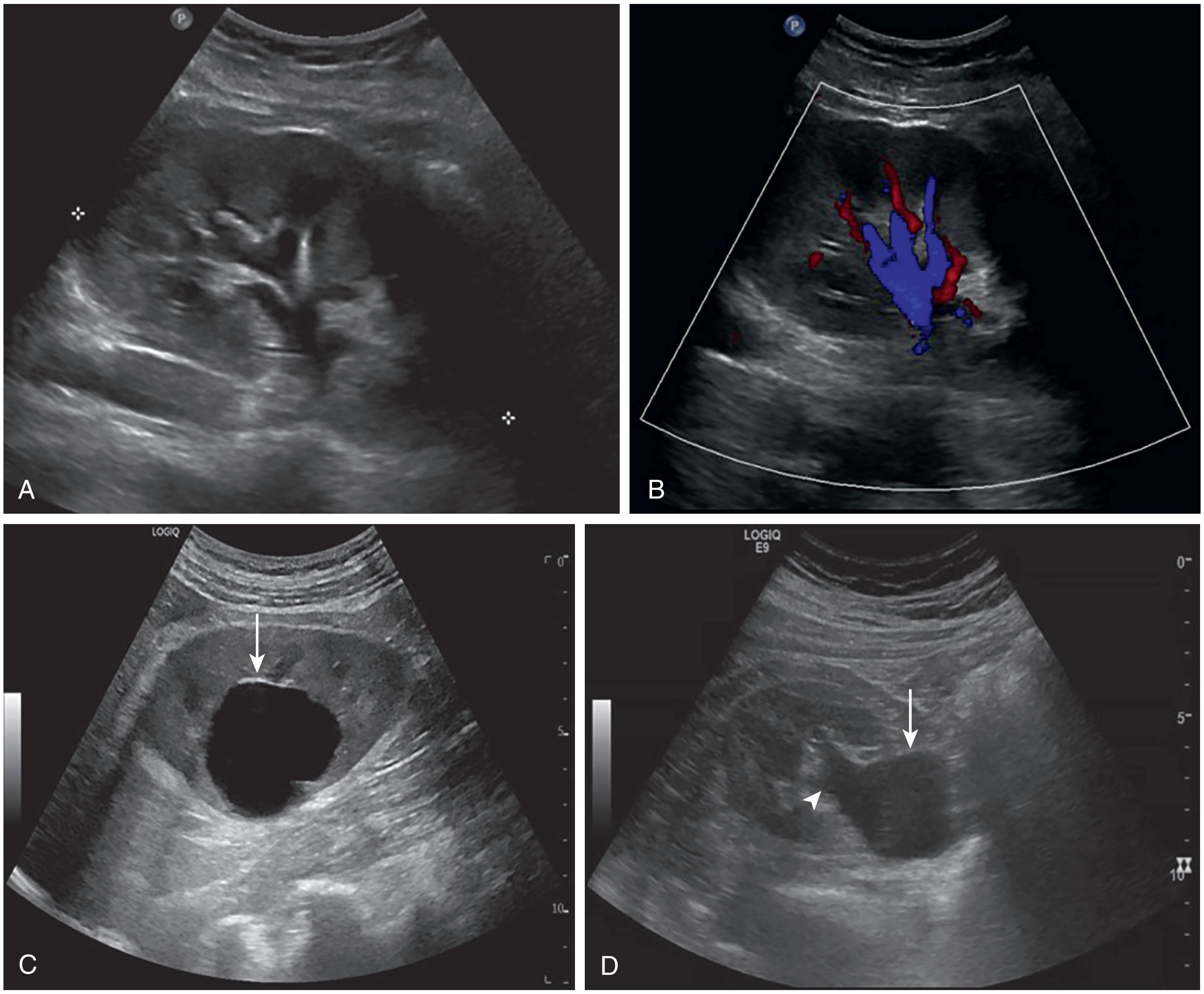


Fig. 26.4 Kidney ultrasound images demonstrating (A) anechoic branching structures in mid kidney concerning for hydronephrosis, but color Doppler (B) demonstrates flow suggesting these are prominent vessels. (C) Parapelvic cyst *(arrow),* (D) extrarenal pelvis (*arrow; arrowhead* demonstrates connection with the collecting system).

system. Parapelvic cysts are another potential mimic due to their anechoic nature and proximity to the renal pelvis. However, hydronephrosis typically appears as a branching structure, whereas parapelvic cysts are noncommunicating, well-circumscribed structures (Fig. 26.4C). Tilting the transducer to visualize the whole thickness of the kidney ensures that the cyst is not communicating with the collecting system unlike hydronephrosis (Video 26.1). An anatomic variant, extrarenal pelvis, may also be confused with hydronephrosis. It is essentially renal pelvis located outside the kidney, demonstrating slight (large in some cases) dilation. It presents as an anechoic roundish structure just outside the renal sinus and, unlike hydronephrosis, doesn't branch into the kidney. The transverse view is particularly useful to demonstrate its connection with the collecting system, which also distinguishes it from a cyst (Fig. 26.4D, Video 26.2). Comparing with any prior available imaging can aid in the identification of all these mimics of hydronephrosis. Notably, the collecting system of a healthy transplanted kidney often exhibits slight dilation. This is likely due to a combination of increased urine production (since it is functioning as the sole kidney) and the loss of ureteral tonicity caused by denervation.

O denotes outline. Because renal masses frequently disrupt the kidney's natural outline and cortical echotexture, it is essential to pay attention to the renal contour and compare it with that of the contralateral kidney. While POCUS is typically not intended for evaluation of masses, identifying an incidental mass can be a valuable opportunity for timely intervention. Fig. 26.5 and Video 26.3 demonstrate a large heterogeneous right renal mass disrupting the kidney architecture, identified during a routine CKD clinic visit. The patient eventually underwent nephrectomy, and histopathology confirmed renal cell carcinoma. Even when there is no obvious mass, any suspicious alteration in kidney outline should prompt a radiology-performed ultrasound examination.

N represents notable abnormalities, such as cysts and stones. On grayscale ultrasound, stones manifest as hyperechoic structures with a shadowing effect behind them (Fig. 26.6A). The ability to detect stones relies on their size and location. According to a study, 73% of stones measuring <3 mm went undetected on ultrasound when compared with computed tomography (CT).[11] Moreover, ultrasound often fails to identify ureteral stones due to bowel gas interference. Essentially, if the clinical question pertains to stone burden, Video 26.1 Parapelvic cyst, long and short axis views *(arrow)*.

Video 26.2 Extrarenal pelvis, long and short axis views *(arrow)*.

Video 26.3. Renal mass, long and short axis views *(arrow)*.

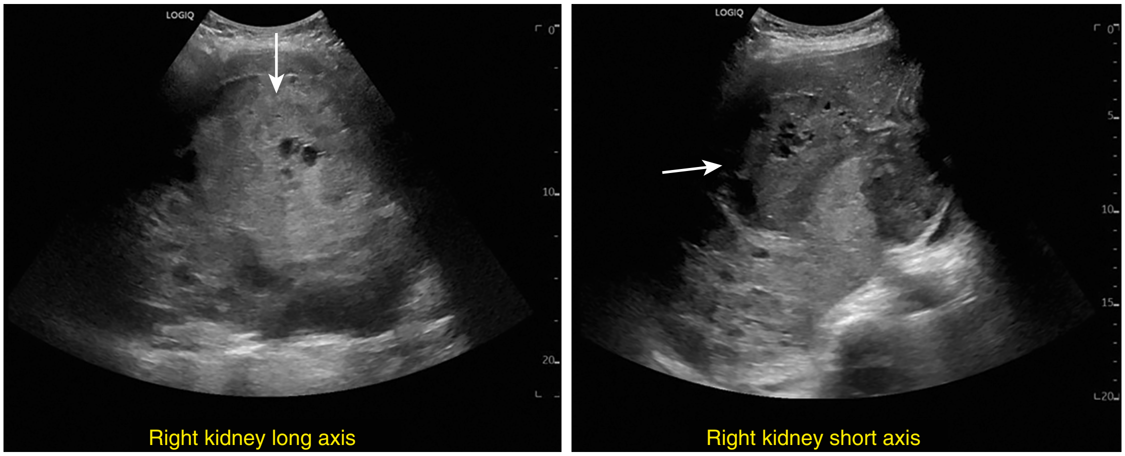


Fig. 26.5 A large, heterogeneous right renal mass *(arrow)* eventually diagnosed as renal cell carcinoma.

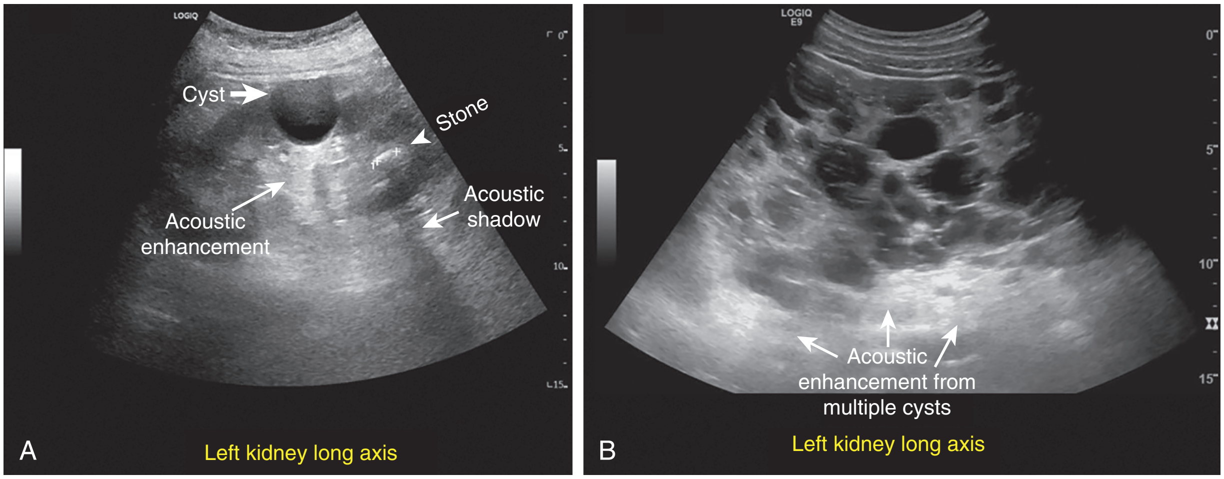


Fig. 26.6 Kidney ultrasound demonstrating (A) a simple cyst *(arrow)* and a stone *(arrowhead),* (B) multiple cysts in a patient with autosomal dominant polycystic kidney disease.

CT is preferred over POCUS. However, POCUS is reliable in detecting urinary obstruction in patients with established stone disease. Cysts appear as anechoic structures on ultrasound with an area of increased brightness behind them, known as acoustic enhancement (see Figs. 26.6A and B). This enhancement is a property of any fluid-filled structure, such as the urinary bladder and can potentially obscure underlying structures. Cysts that are completely anechoic without any internal echoes are called simple cysts whereas those with septations, calcifications, or solid components are termed complex cysts. Except for cysts with only a few thin septations, it is advisable to obtain follow-up imaging, preferably a contrast-enhanced CT, to guide further management when a complex cyst is detected (Fig. 26.7 and Video 26.4).

D stands for Doppler and underscores the importance of using color Doppler to differentiate between urine and blood, such as in distinguishing hydronephrosis from prominent vasculature, as previously mentioned. It also enables the potential identification of renal infarcts, which can manifest as areas lacking blood flow in a kidney that appears, otherwise normal on grayscale images. In addition to color Doppler, pulsed wave Doppler can be employed to evaluate renal venous congestion in cases of fluid overload, a topic we delve into later in this chapter. The final S in the checklist refers to the surroundings and signifies identification of perinephric collections, especially in the assessment of renal allografts (e.g., lymphocele, urinoma, and hematoma). Depending on the clinical context, POCUS users should also note the presence of ascites or blood (e.g., postbiopsy hematoma) around the kidney (Fig. 26.8, Video 26.5).

Urinary bladder POCUS must always be performed in conjunction with kidney when evaluating a patient with AKI. The urinary bladder appears as a well-defined, anechoic area in the pelvis when it contains urine. In cases where the bladder has been emptied using a Foley catheter, a cystic structure corresponding to the fluid-filled Foley balloon is seen enclosed by the bladder wall (Fig. 26.9A and 26.9B). A quick bladder POCUS can identify abnormalities such as an enlarged prostate, obstructed Foley catheter, clots, and even tumors (Fig. 26.9C–26.9E). Furthermore, bladder POCUS aids in distinguishing pelvic ascites from the urinary bladder, a task that automated bladder scanners often fail to accomplish. Pelvic ascites presents as an ill-defined anechoic area connected to the rest of the peritoneal cavity, unlike the distinctly outlined urinary bladder located within the pelvis

Video 26.4 Complex cyst *(arrow)* in the left kidney upper pole with thick septations and internal flow on color Doppler.

Video 26.5 (A) Lymphocele adjacent to a renal allograft demonstrating faint septations (expected) and (B) ascites in between right native kidney and the liver. Images are from different patients.

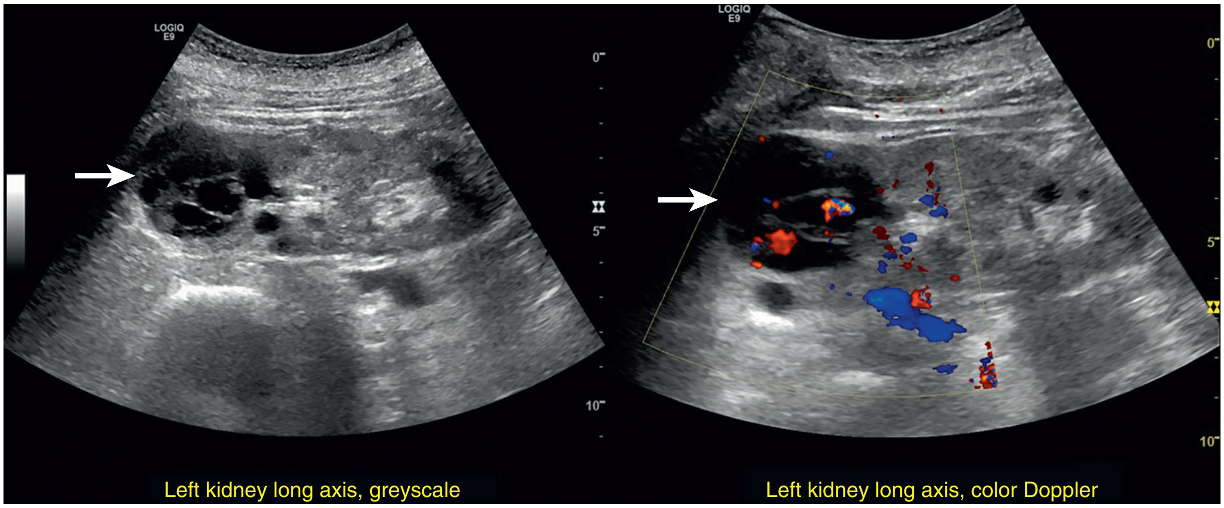


Fig. 26.7 Complex cyst *(arrow)* in the left kidney upper pole with thick septations and internal flow on color Doppler concerning for malignancy.

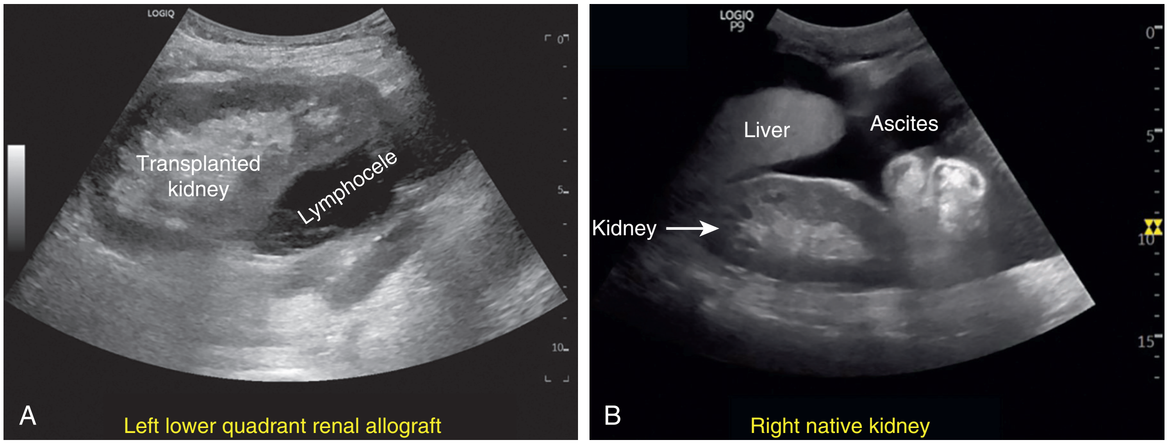


Fig. 26.8 Ultrasound images demonstrating (A) lymphocele adjacent to a renal allograft, (B) ascites in between right native kidney and the liver. Images are from different patients.

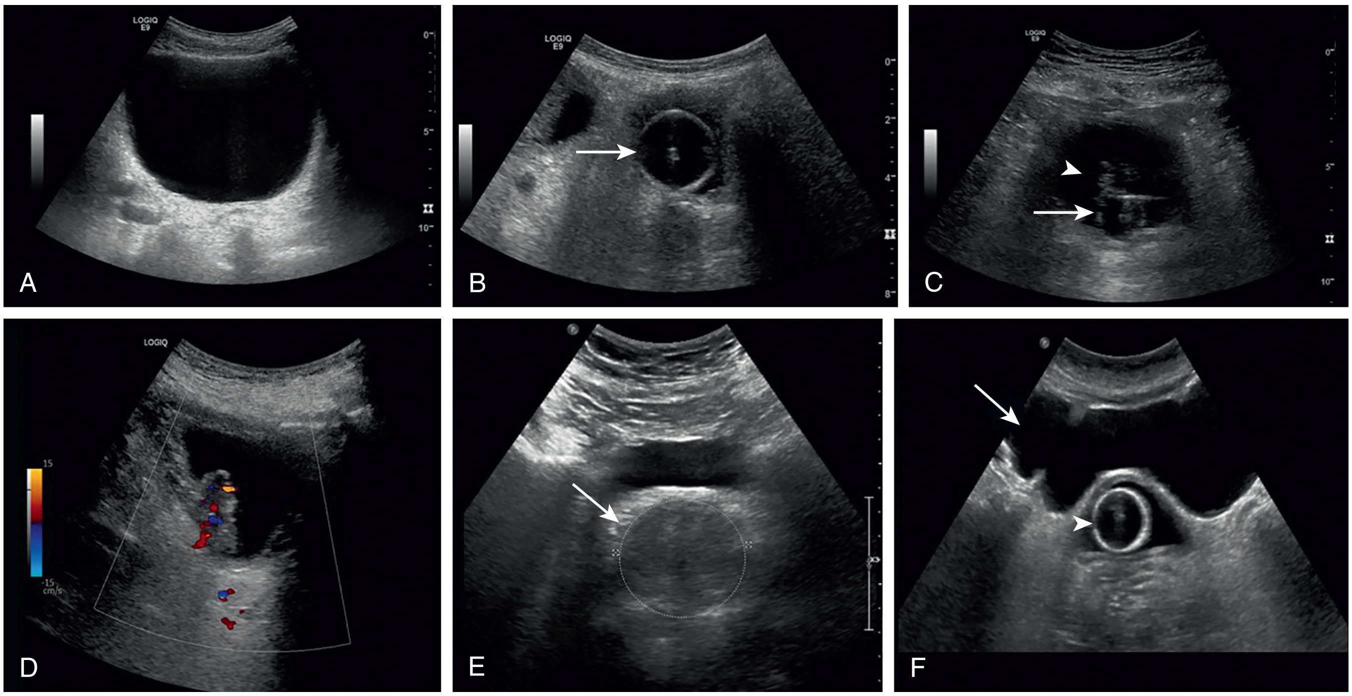


Fig. 26.9 Urinary bladder ultrasound images demonstrating (A) normal urine-filled bladder, (B) decompressed bladder with Foley catheter (*arrow* points to Foley balloon), (C) bladder containing urine despite the presence of Foley catheter (*arrow* points to Foley balloon, arrowhead points to debris), (D) bladder mass with internal flow on color Doppler, (E) enlarged prostate *(arrow),* and (F) pelvic ascites *(arrow)* anterior to decompressed urinary bladder (*arrowhead* points to Foley balloon).

(Fig. 26.9F). In females, the uterus suspended in pelvic ascites is often seen and should not be mistaken for a mass.[12]

HEMODYNAMIC ASSESSMENT

Accurate assessment of a patient's volume status is a common challenge for nephrologists managing complex fluid and electrolyte disorders. The difficulty arises from the fact that no single diagnostic method, including physical examination and laboratory testing, is entirely foolproof in this regard.[13] While imaging tests such as chest radiography and echocardiography help, repeatedly performing these tests to monitor response to treatment isn't practical. Additionally, it is not cost-effective to redo an entire study simply to address a focused clinical question or evaluate a single parameter, like presence or absence of pleural effusion or renal venous congestion. This is where POCUS serves as a valuable component of the bedside clinical assessment. Goal-directed POCUS of the pump (focused cardiac ultrasound [FoCUS]), pipes (inferior vena cava [IVC] ultrasound and venous Doppler), and leaks (evaluation of extravascular lung water and ascites) offers helpful insights into a patient's hemodynamics when interpreted in the appropriate clinical context.[14] While mentioned separately here, IVC ultrasound is generally considered a component of FoCUS. On the other hand, heavy reliance on isolated organ POCUS (for instance, exclusively examining the lungs or just the IVC) can result in diagnostic errors and potential patient mismanagement. It is important to emphasize that POCUS doesn't provide a direct measurement of the blood volume; instead, it aids in gauging the functional status of the hemodynamic circuit. Therefore the term "hemodynamic assessment" is preferred over "volume status" assessment.

FOCUSED CARDIAC ULTRASOUND

FoCUS is a limited ultrasound examination that involves obtaining a specific set of high-yield cardiac views to assist clinical decision making. Noncardiologist physicians, often with a few weeks of training, can perform this assessment.[15] Fig. 26.10 illustrates the basic cardiac views acquired during FoCUS, along with the corresponding transducer positions.

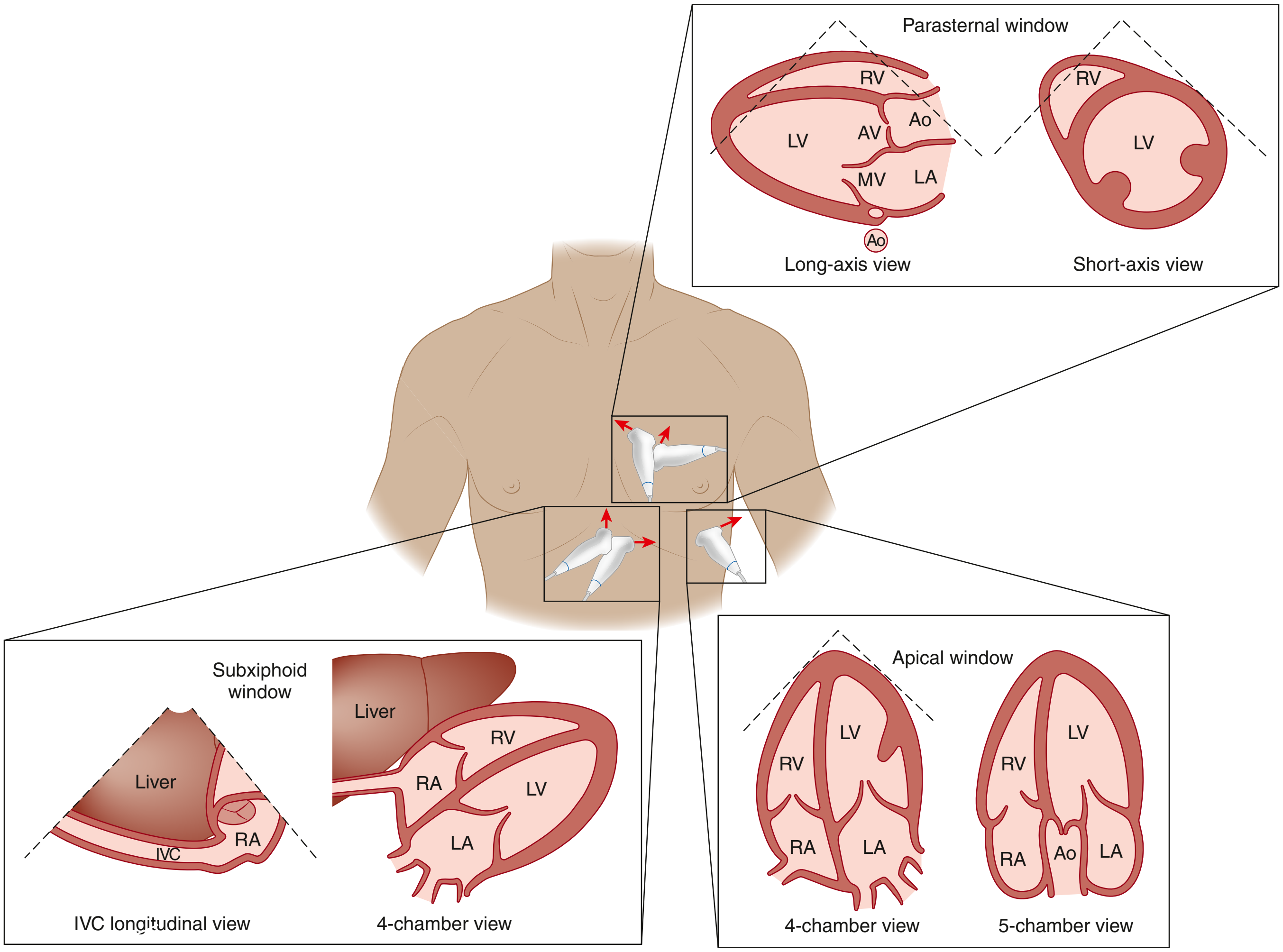


Fig. 26.10 Illustrations of the transducer positions for obtaining basic cardiac views, along with the corresponding images. *Arrows* indicate the direction of the transducer orientation marker. *Ao,* Aorta; *IVC,* inferior vena cava; *LA,* left atrium; *LV,* left ventricle; *MV,* mitral valve; *RA,* right atrium; *RV,* right ventricle.

While FoCUS might alleviate the need for consultative echocardiography in specific cases, it cannot be viewed as a substitute. The latter involves a comprehensive examination, documenting numerous predetermined parameters, and demands more intensive training.

Similar to the SECONDS checklist for kidney ultrasound interpretation, the "5 Es" is a helpful mnemonic for ensuring a systematic approach to interpreting FoCUS findings. These 5 Es denote **E**jection (left ventricular [LV] systolic function), **E**ffusion (pericardial), **E**quality (relative cardiac chamber size), **E**ntrance (to the heart, i.e., IVC), and **E**xit (from the heart, i.e., aorta).[16]

Measuring LV ejection fraction can be time consuming owing to the necessity of tracing the endocardial border in two different views. In the point-of-care scenario, LV systolic function is often qualitatively categorized as reduced, normal, or hyperdynamic through visual assessment (eyeballing) of the ventricular wall and mitral valve movements. Parasternal long- and short-axis views (PLAX, PSAX) are typically used for this assessment. This information can be used to determine whether a patient with hypotension and renal dysfunction would benefit from diuretic therapy, intravenous fluids, or vasoactive agents. In general, it is expected that the walls of the LV thicken and will come closer together by one-third or more during systole. In patients with reduced LV systolic function, both inward motion and wall thickening are impaired. Conversely, a hyperdynamic LV, where the ventricular walls and papillary muscles in the PSAX view nearly touch at end-systole, suggests volume depletion or a reduced afterload state (e.g., vasoplegia). In addition, normally, the anterior leaflet of the mitral valve almost touches the interventricular septum in the PLAX view during diastole. If the ejection fraction is reduced, resulting in increased end-systolic volume, blood flow across the mitral valve decreases during diastole and does not open fully. This increases the distance between the septum and the anterior leaflet (Video 26.6). M-mode across the mitral valve leaflets can also be used to document this. More than 7-mm distance between the peak of the anterior mitral valve tracing during early passive diastolic filling (E-point) and the interventricular septum is considered a marker of significant LV systolic dysfunction (LVEF <30%).[17] This measurement is referred to as the E-point septal separation (EPSS) (Fig. 26.11) On a note of caution, aortic regurgitation and mitral valve stenosis can reduce mitral valve excursion in diastole, resulting in a low EPSS despite preserved ejection fraction. Color Doppler helps to demonstrate aortic regurgitation and structural alterations such as calcified leaflets can provide a clue to mitral valve stenosis.

Pericardial effusion appears as an anechoic space between the two layers of pericardium. While it is generally identifiable in all standard FoCUS views (Fig. 26.12), the PLAX and subxiphoid views offer easier visualization. A diameter of <1 cm between the pericardial layers during diastole is generally considered as mild effusion, 1 to 2 cm as moderate, and >2 cm as large, though some regional variation is often present. Of note, in the parasternal views, there's a potential for confusion between a left pleural effusion and pericardial effusion because the pleural space is immediately posterior to the pericardium. The descending thoracic aorta serves as an anatomic landmark to differentiate them. Pericardial effusion is located anterior to the aorta, whereas the pleural effusion is seen posteriorly or commences at the same level (Video 26.7). Whenever a pericardial effusion is identified, it is important to look for right-sided chamber collapse in addition to distended IVC, which is suggestive of tamponade physiology. Fig. 26.13 and Video 26.8 demonstrate diastolic right ventricular (RV) outflow tract collapse, obtained from a dialysis patient. Notably, patients experiencing subacute tamponade are frequently hypertensive rather than hypotensive.[18] Aggressive ultrafiltration or diuretic therapy aimed at hypertension without the knowledge of presence of effusion can induce tamponade physiology by lowering intracardiac pressure in relation to pericardial pressure.

Equality primarily denotes the relative size of RV and LV. In the apical four-chamber view, a normal RV is smaller than

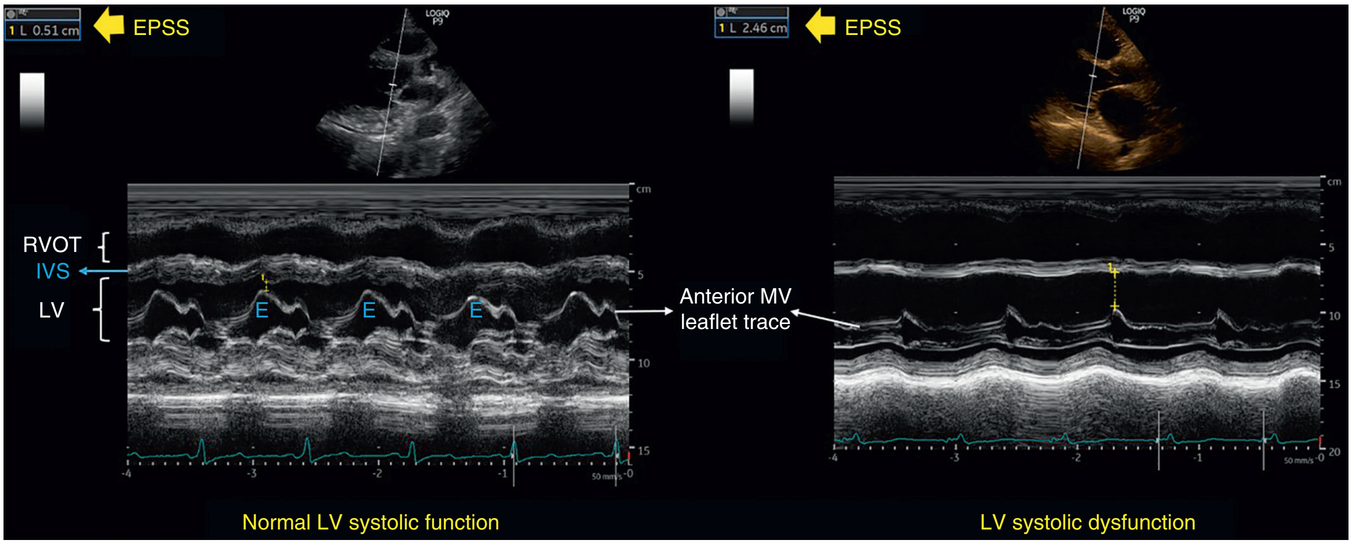


Fig. 26.11 E-point septal separation (EPSS) using M-mode demonstrating normal and severely reduced left ventricular systolic function. *E,* Early diastolic wave of the mitral valve excursion; *IVS,* interventricular septum; *LV,* left ventricle; *MV,* mitral valve; *RVOT,* right ventricular outflow tract.

Video 26.6 Qualitative assessment of the left ventricular systolic function on the parasternal long axis view. *Ao,* ascending aorta; *LA,* left atrium; *LV,* left ventricle; *RVOT,* right ventricular outflow tract.

Video 26.7 Distinction between pericardial and pleural effusions on the parasternal long and short axis views.

Video 26.8 Parasternal long and short axis views demonstrating pericardial effusion and diastolic collapse of the right ventricular outflow tract *(arrow).*

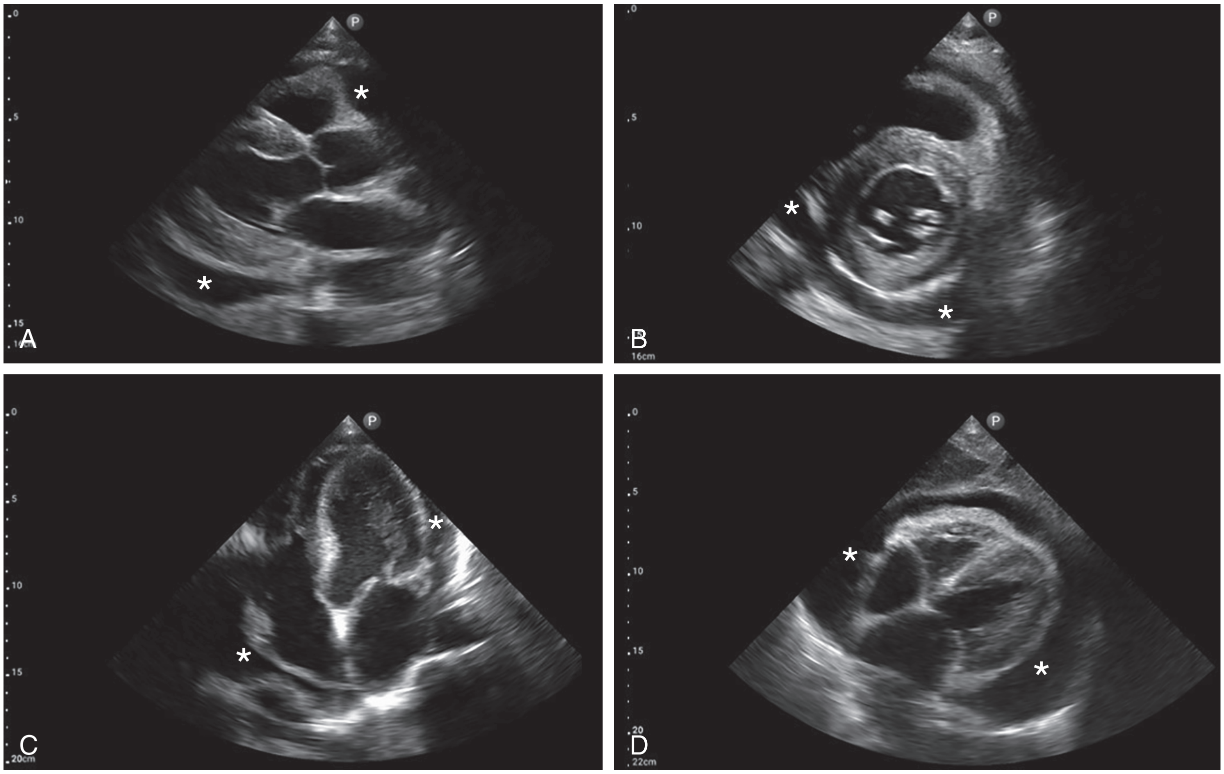


Fig. 26.12 Pericardial effusion *(asterisks)* seen on the (A) parasternal long axis view, (B) parasternal short axis view, (C) apical 4-chamber view, and (D) subxiphoid 4-chamber view.

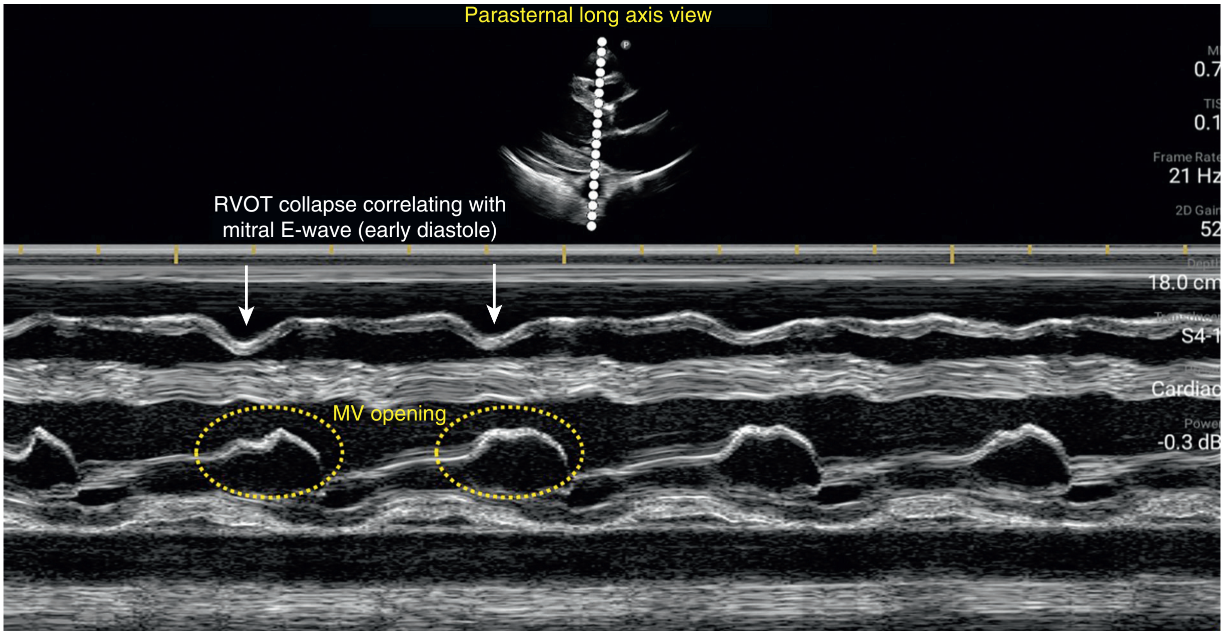


Fig. 26.13 M-mode image demonstrating diastolic right ventricular outflow tract (RVOT) collapse in the parasternal long axis view. *MV*, Mitral valve.

two-thirds the size of the LV. When the RV is slightly enlarged due to pressure or volume overload, its size is just over two-thirds of the LV, and severe RV dilation is characterized by it being larger than the LV. In the PSAX view, RV dilation results in the flattening of the interventricular septum, causing the LV to take on a D-shaped appearance instead of its usual round shape (the D-sign) (**Fig. 26.14 and Video 26.9**). Detecting RV enlargement is crucial in three nephrology-related scenarios. Firstly, in a patient with volume overload, often in conjunction with tricuspid regurgitation, hypotension can be a consequence of reduced stroke volume due to LV compression. In such cases, empiric administration of intravenous fluids can be detrimental. Secondly, in a patient with pulmonary hypertension and RV

Video 26.9 Severe right ventricular enlargement as seen on the apical four-chamber and parasternal short axis views.

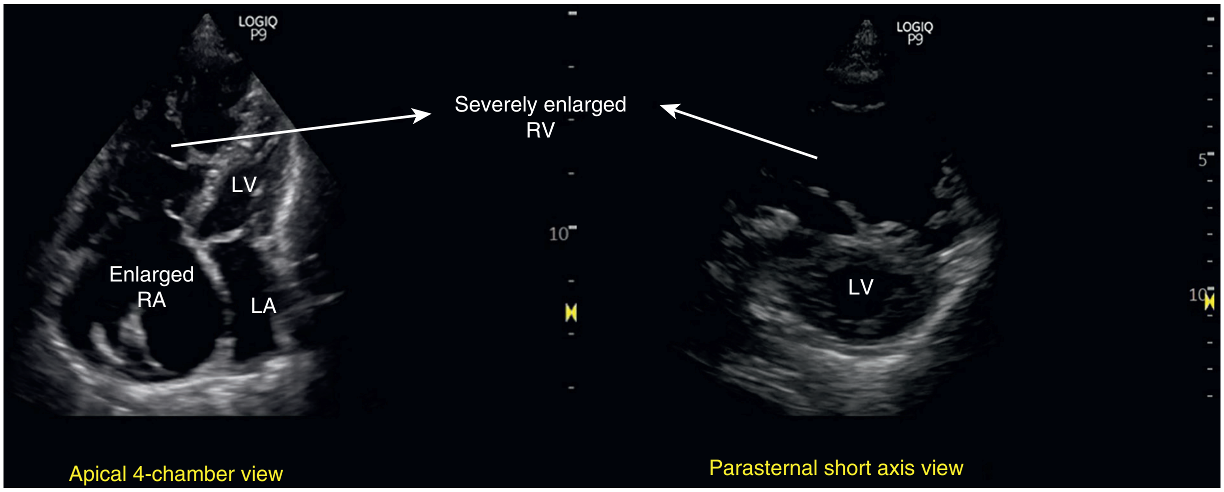
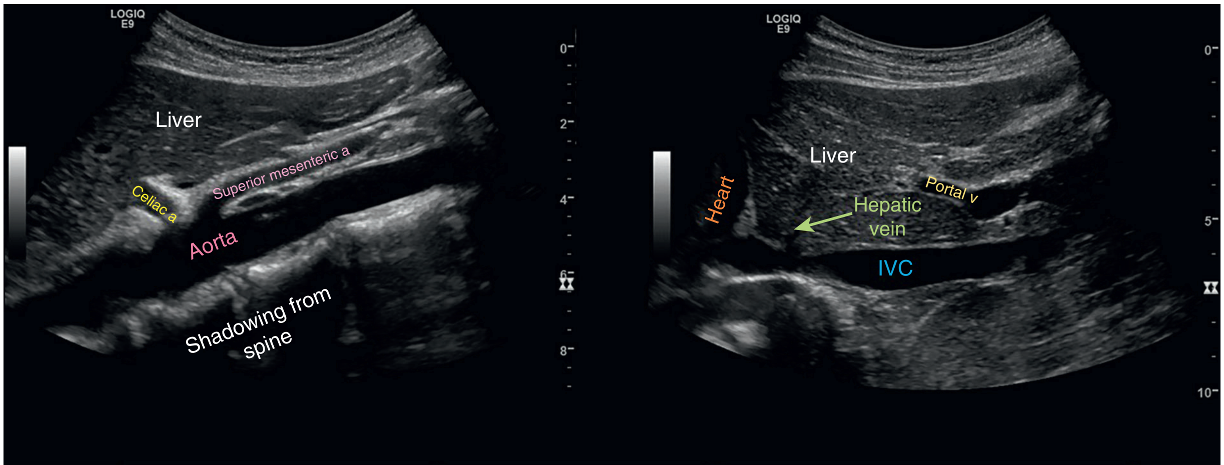


Fig. 26.14 Severe right ventricular *(RV)* enlargement as seen on the apical four-chamber and parasternal short axis views. Note interventricular septal flattening on the short axis view. *LA,* Left atrium, *LV,* left ventricle; *RA,* right atrium.



IVC maximal diameter	Collapsibility with a sniff	Estimated right atrial pressure
Less than or equal to 2.1 cm	More than 50%	3 mm Hg
More than 2.1 cm	Less than 50%	15 mm Hg
Less than 2.1 cm	Less than 50%	8 mm Hg
More than 2.1 cm	More than 50%	8 mm Hg

Fig. 26.15 *Left,* Aorta versus inferior vena cava *(IVC)* on ultrasound—aorta is separated from the liver. It is in front of the vertebral bodies and has anterior branches outside the liver (i.e., celiac, and superior mesenteric arteries). It cannot be traced down to the heart as it is continuous with the thoracic aorta. The IVC is attached to the liver, and the hepatic vein can be seen draining into it and entering the right atrium at the level of the diaphragm. It does not have major anterior tributaries outside the liver. *Right,* IVC parameters used to estimate right atrial pressure in spontaneously breathing patients.

failure, diuretic therapy may diminish preload and cause AKI. Identification of the dysfunctional RV with a plethoric IVC should prompt expert consultation and consideration of pulmonary vasodilator therapy for these individuals. Lastly, in patients who are predisposed to thromboembolic events, such as those with nephrotic syndrome presenting with shortness of breath, RV enlargement, and reduced longitudinal motion (evaluated by observing the movement of the tricuspid annulus toward the apex in apical four-chamber view) could be a clue to a hemodynamically significant pulmonary embolism.

Entrance denotes evaluation of the maximal anteroposterior diameter and collapsibility of the IVC to estimate right atrial pressure (RAP) (Video 26.10); elevated RAP is the prime driver of systemic venous congestion and congestive nephropathy.[19] Fig. 26.15 lists the cutoffs of the parameters used to estimate RAP in spontaneously breathing patients, as well as key differences between the imaging characteristics of aorta and IVC, which are often confused for one another by novice POCUS users. Asking the patient to sniff is a way of standardizing respiratory effort. As the strength of breath varies in hospitalized

Video 26.10 Normal and plethoric inferior vena cavae.

patients, this must be interpreted with caution. In addition, IVC ultrasound is not reliable to assess RAP in mechanically ventilated patients. While IVC ultrasound serves as a useful indicator of *fluid tolerance*, signaling a point to halt fluid administration when the IVC is plethoric, it should not be solely relied upon to determine *fluid responsiveness* (i.e., an increase in cardiac output in response to fluids). Whenever feasible, stroke volume assessment must be performed, as described later. Furthermore, it is important to understand that a small, collapsible IVC is not a definitive indicator of volume depletion. It could indicate either volume depletion or a vasodilatory state with hypervolemia (e.g., cirrhosis) or euvolemic state. Consequently, it should not be employed in isolation to differentiate between hypovolemic and euvolemic hyponatremia or to determine the necessity for fluids in a patient with cirrhosis and AKI. Likewise, a plethoric IVC does not always warrant fluid removal. It is essential to consider concurrent conditions like pericardial effusion and pulmonary hypertension to develop an appropriate management strategy.

In emergency medicine, the term "exit" was employed to refer to the assessment of aortic dilation and dissection in patients presenting with undifferentiated hypotension.[16] For nephrologists, it has evolved to signify something more pertinent—the estimation of stroke volume and cardiac output using pulsed wave Doppler evaluation of the LV outflow tract flow.[14] Fig. 26.16 illustrates this calculation and the underlying concept. In point-of-care settings, LV outflow tract velocity time integral (VTI) is used as a surrogate for stroke volume (normal: ~18–22 cm) to quickly gauge the response to a fluid bolus. It is also useful to distinguish high cardiac output states from volume depletion as both these conditions present with hyperdynamic LV and a small collapsible IVC.[20-22]

In addition to 5 Es, FoCUS can be used to assess gross valvular lesions such as tricuspid regurgitation in a patient with fluid overload, mitral annular calcification in a patient undergoing dialysis, etc. Furthermore, nephrologists with advanced training can evaluate LV diastolic function and estimate pulmonary artery pressures, enabling them to make more informed decisions regarding patient management.

LUNG ULTRASOUND

Lung ultrasound has gained prominence as a valuable bedside tool due to an abundance of literature highlighting its diagnostic superiority compared with auscultation and prognostic significance, particularly among patients with heart failure and those undergoing renal replacement therapy.[23,24] In fact, it has been shown that lung ultrasound is more sensitive than chest radiography for detection of cardiogenic pulmonary edema.[25,26] Lung ultrasound enables nephrologists to promptly address common clinical questions, such as "Does this dialysis patient have pulmonary congestion?" "Would additional ultrafiltration benefit this patient with AKI receiving dialysis?", and "Is this patient's shortness of breath attributable to pleural effusion, pulmonary edema, or another cause?" Moreover, it is the least technically demanding among the sonographic applications mentioned in this chapter and can be performed using entry-level ultrasound devices.

Because lung ultrasound provides information only at the transducer's location, it is essential to scan multiple sites across the thorax to ensure accurate detection and characterization of pathology. Various scanning protocols have been described including a 28-zone approach (16 on the right and 12 on the left),[27] an 8-zone approach (4 on each side),[28] a 6-zone approach (3 on each side),[29] and a 4-zone approach (2 on each side).[30] Except for the 28-zone protocol, which involves scanning each intercostal space, a *zone* usually includes multiple intercostal spaces depending on the transducer used. The premise is that scanning fewer zones may suffice when the pathology is diffuse and the patient is symptomatic while scanning more zones becomes necessary when the patient is asymptomatic and/or the

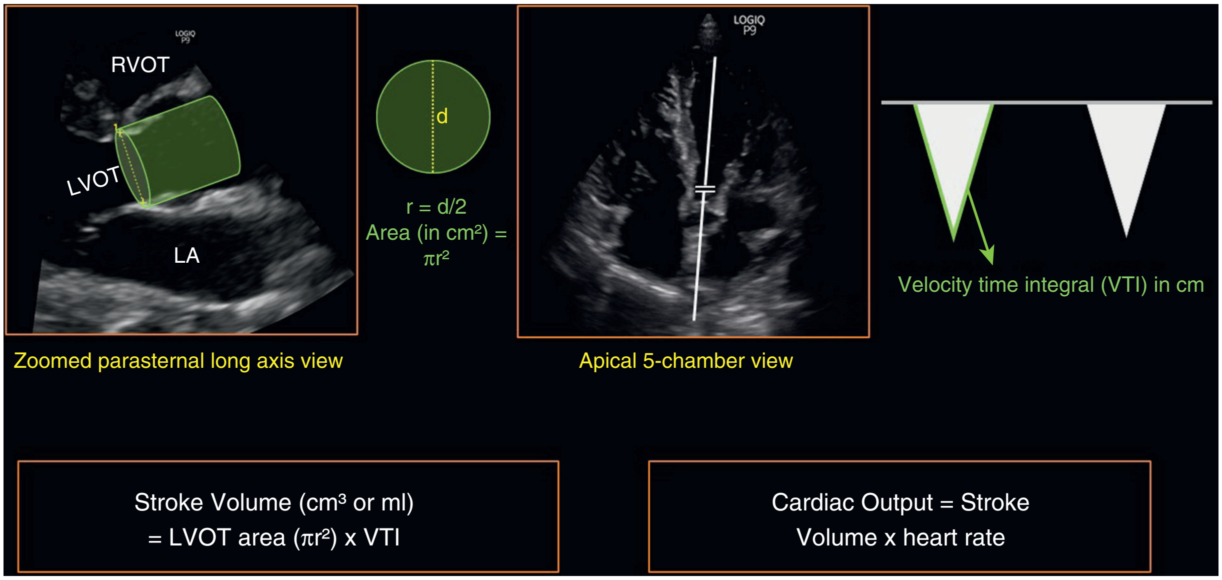


Fig. 26.16 Stroke volume is the product of the cross-sectional area of left ventricular outflow tract *(LVOT)* and velocity time integral *(VTI).* LVOT area is derived from the LVOT diameter in zoomed parasternal view using the formula πr2, and LVOT VTI is obtained by tracing the envelope of the pulsed wave Doppler spectrum of systolic flow in the apical five-chamber view.

pathology is unevenly distributed. As a result, most studies in hemodialysis patients aimed at detecting subclinical lung congestion have used the comprehensive 28-zone protocol. However, recent research has shown that abbreviated protocols still retain prognostic significance.[31,32] In routine practice, an 8-zone protocol is typically employed, involving the scanning of two anterior and two lateral zones on each hemithorax (Fig. 26.17). The inclusion of lateral zones is particularly beneficial in identifying pulmonary congestion in dependent areas and pleural effusions.

Lung tissue is not visible on ultrasound beyond the pleural interface unless it is consolidated or atelectatic. This is because the underlying air reflects ultrasound waves and creates artifacts. Normal lung demonstrates equidistantly placed horizontal hyperechoic artifacts referred to as "A-lines" following the bright shimmering pleural line where the shimmer corresponds to pleural sliding. These artifacts are a result of ultrasound reverberation originating from the pleural interface. Conversely, vertical hyperechoic artifacts known as "B-lines" appear when there is a reduction in lung air content caused by interstitial thickening, often due to the presence of fluid (Fig. 26.17, Video 26.11). In other words, B-lines generally indicate increased extravascular lung water. B-lines originate at the pleural line, extend to the edge of the ultrasound image without fading, and move synchronously with lung sliding. A positive "B-line region" is characterized by at least three B-lines observed in a longitudinal plane between two ribs. Having at least two positive regions on both sides indicates the presence of "interstitial syndrome," which is indicative of a diffuse process such as pulmonary edema.[33] When there is a significant loss of lung aeration (as in transition from interstitial to alveolar edema), B-lines can merge, completely obscuring the A-lines and causing the lung to appear uniformly white. When discrete, the total number of B-lines can be counted to quantify the severity of congestion. In a study including 392 hemodialysis patients, individuals with severe lung congestion (>60 B-lines on a 28-zone ultrasound) experienced a higher incidence of all-cause mortality, as well as fatal and nonfatal cardiac events when compared with those with

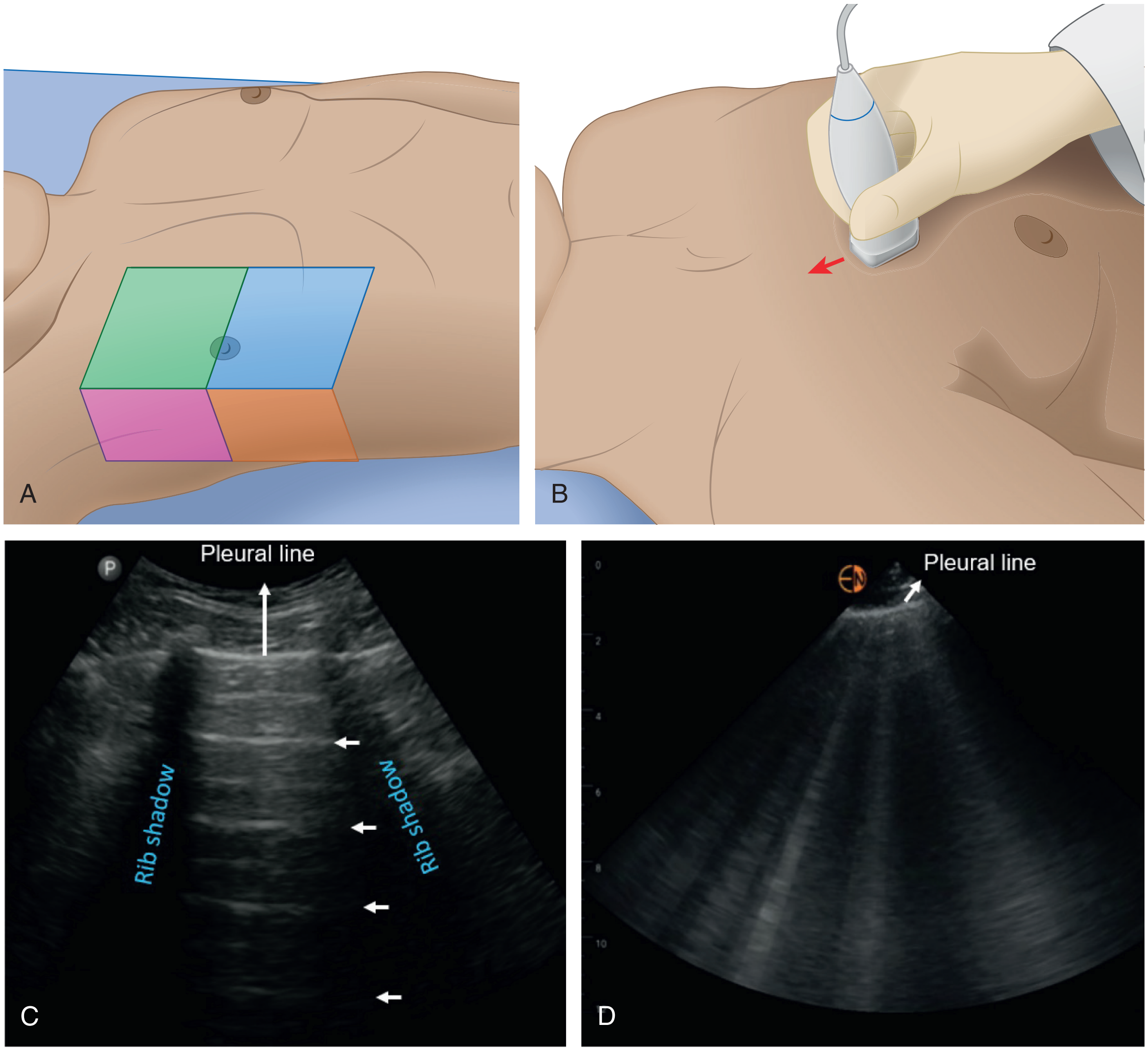


Fig. 26.17 [Upper panel] (A) Illustrations depicting commonly used lung ultrasound zones, with two anterior and two lateral zones in each hemithorax. Depending on the transducer used, one zone may encompass more than one intercostal space. (B) Illustration demonstrating the proper alignment of the transducer during lung ultrasound, ensuring the transducer surface is perpendicular to the chest wall while considering its curvature. Lower panel: (C) Horizontal A-lines (normal finding). (D) Vertical B-lines observed in a patient with pulmonary congestion.

Video 26.11 Lung ultrasound images demonstrating A- and B-lines.

moderate congestion (15–60 B-lines). Further, moderate congestion conferred worse prognosis compared with mild congestion (fewer than 15 B-lines).[34] The number of B-lines dynamically decreases with volume removal, making them a potential target for monitoring the effectiveness of decongestive therapy. While the concept of adjusting dry weight based on B-line guidance appears appealing in patients on maintenance dialysis, such an approach may overlook the presence of right-sided congestion. Of note, pulmonary hypertension and RV dysfunction are highly prevalent among dialysis patients, especially those with brachial arteriovenous fistulas, and associated with a poorer prognosis.[35,36] Therefore an optimal approach should consider multiple sonographic parameters to assess the entire hemodynamic circuit interpreted in conjunction with clinical and laboratory data to have a meaningful impact on long-term outcomes. Not unsurprisingly, a large clinical trial known as the Lung Water by Ultrasound-Guided Treatment in Hemodialysis Patients (LUST) did not demonstrate a statistically significant difference in the primary composite endpoint of all-cause death, nonfatal myocardial infarction, and decompensated heart failure, in hemodialysis patients managed with a B-line–guided protocol. However, a post hoc analysis that examined recurrent episodes of acute decompensated heart failure revealed a hazard ratio of 0.37 for those in the interventional arm, indicating a reduction in the risk of recurrent cardiovascular events.[37] Additionally, in a LUST substudy involving 71 *apparently euvolemic* patients with hypertension, lung ultrasound-guided ultrafiltration resulted in significantly better control of ambulatory blood pressure compared with controls.[38] An important technical limitation is that B-lines are not exclusive to cardiogenic pulmonary edema; they can also appear in patients with diffuse parenchymal lung diseases with interstitial thickening, such as pulmonary fibrosis, and alveolar filling processes like pneumonia and acute respiratory distress syndrome (ARDS). Findings such as pleural line irregularities, reduced sliding, spared areas, and subpleural consolidations suggest pneumogenic over cardiogenic etiology, but the distinction is not always straightforward (e.g., a patient with stigmata of recent COVID-19 infection with superimposed fluid overload). Assessment of LV filling pressures by Doppler echocardiography aids in such cases.

Lung ultrasound also exhibits excellent test characteristics for the diagnosis of pleural effusion and pneumonia outperforming auscultation and chest radiography.[39,40] Pleural effusion appears as an anechoic space above the diaphragm, and the atelectatic lung is often seen floating within (Video 26.12). Pneumonia shares similarities with atelectasis in terms of tissue characteristics, often resembling the appearance of the liver (hepatization). Clinical and laboratory stigmata of infection, presence of blood flow within the affected lung, dynamic air bronchograms represented by moving bright spots in the airways, and the identification of septations within the surrounding effusion, favor a diagnosis of pneumonia over compressive atelectasis.

INTERNAL JUGULAR VEIN ULTRASOUND

Traditionally, examining jugular venous pulsations has been a method for estimating RAP in patients with heart failure. However, accurate assessment needs expertise and can be challenging, especially in patients with obesity resulting in low sensitivity and poor reproducibility.[41] The internal jugular vein (IJV) can be readily identified with ultrasound imaging, significantly enhancing its detectability compared with visual examination alone. Given the superficial nature of the IJV, a high-frequency linear transducer is used to image it (Fig. 26.18). There are a few different ways to estimate RAP using IJV POCUS. Evaluation of changes in IJV diameter and/or area during the Valsalva maneuver is a well-described method with good interoperator reproducibility[42] (Fig. 26.19). In a study involving 342 patients with heart failure, a ratio between maximum IJV diameter during Valsalva maneuver and diameter at rest <4 (i.e., less dilation with Valsalva) was independently associated with all-cause mortality or heart failure hospitalization at 3 months.[43] A similar technique assessing the Valsalva maneuver-induced changes in the IJV cross-sectional area found that a <66% change in area was a good predictor of a RAP ≥12 and was associated with increased readmissions for heart failure.[44]

With respect to other methods, in a study involving 100 patients undergoing right heart catheterization, Wang and colleagues[45] performed ultrasound evaluation of the IJV with the head of the bed at 30 to 45 degrees beginning with the transducer located at the neck above the clavicle. Once the IJV is identified, the vessel is followed cranially until the point of collapse is located. The point of IJV collapse is the point at which the IJV is smaller than the adjacent carotid artery throughout the entirety of the respiratory cycle. The vertical height of the IJV collapse point is then measured from the sternal angle and expressed in centimeters and 5 cm are added (accounting for the depth of the RA). This ultrasound jugular venous pulse showed excellent correlation with RAP measurements and accurately predicted a RAP ≥10 mm Hg (area under the curve: 0.84, 95% confidence interval: 0.76–0.92). A qualitative method based on identification of the IJV collapse point in reference to neck position also showed good correlation with RAP[45] (Fig. 26.20). Recently, RAP estimation using height of the IJV collapse together with measured RA depth (instead of assuming to be 5 cm) using PLAX cardiac view has been shown to accurately estimate RAP within 3 mm Hg in 67% of patients.[46] Of note, IJV POCUS can be especially relevant for the evaluation of congestion in conditions with altered IVC anatomy such as cirrhosis. A study in patients with cirrhosis showed that IJV respiratory collapse assessed by ultrasound was superior to IVC ultrasound for estimating RAP.[47]

DOPPLER-ASSISTED VENOUS CONGESTION ASSESSMENT

Hemodynamically mediated organ injury in the setting of heart failure can occur secondary to reduced cardiac output or increased RAP. Invasive hemodynamic measurements in patients with acute decompensated heart failure have demonstrated that venous congestion, rather than decreased cardiac output, is the most important factor driving organ dysfunction.[48] Thus monitoring venous congestion at the bedside can provide useful information regarding the possibility of congestive organ injury.

Video 26.12 Lung ultrasound images demonstrating pleural effusion. *Arrowhead* indicates atelectatic lung. *Asterisks* indicate shadows from thoracic vertebrae ('*spine sign*'), which are not visualized in the absence of effusion or consolidation (obscured by air in the lung).

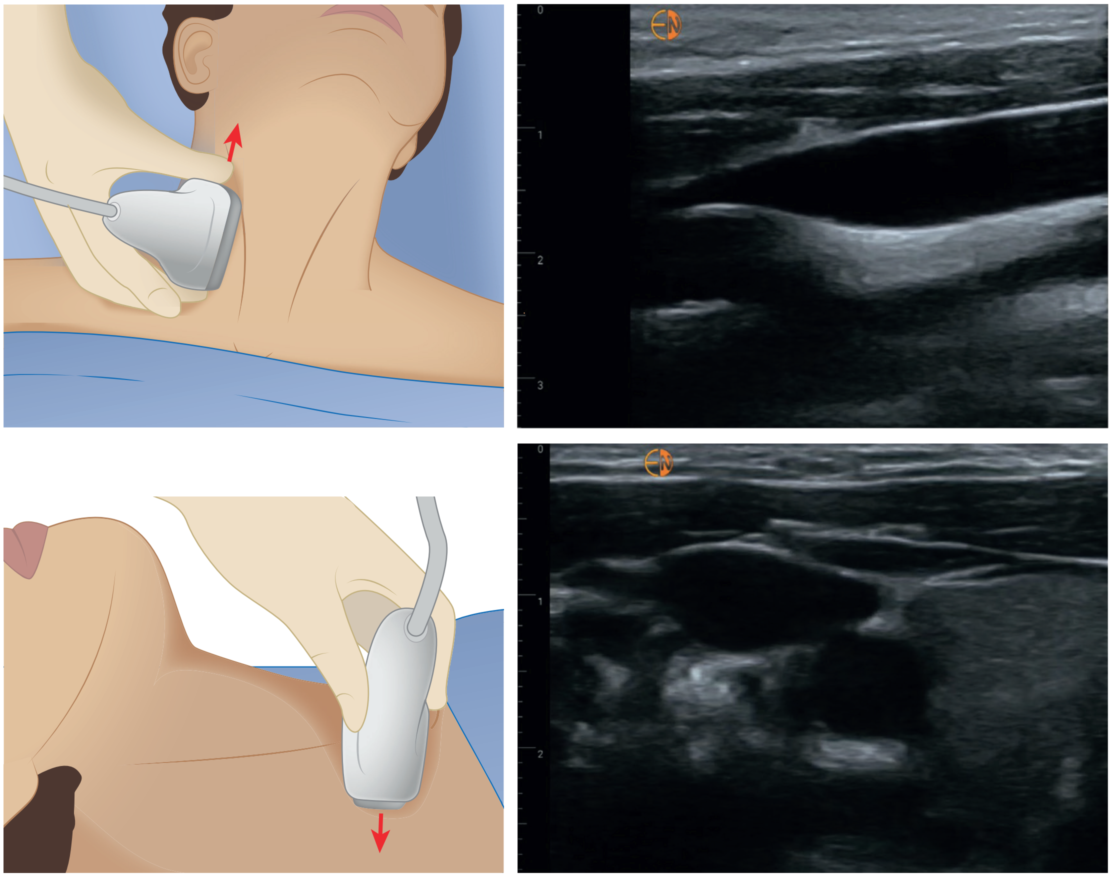


Fig. 26.18 Left panels: Illustrations depicting the transducer position for obtaining long and short-axis views of the internal jugular vein. *Arrows* indicate the direction of the transducer orientation marker. It's important to keep the patient's head in a neutral position as much as possible to prevent muscle tension on the vessel. Right panels: Corresponding sonographic images.

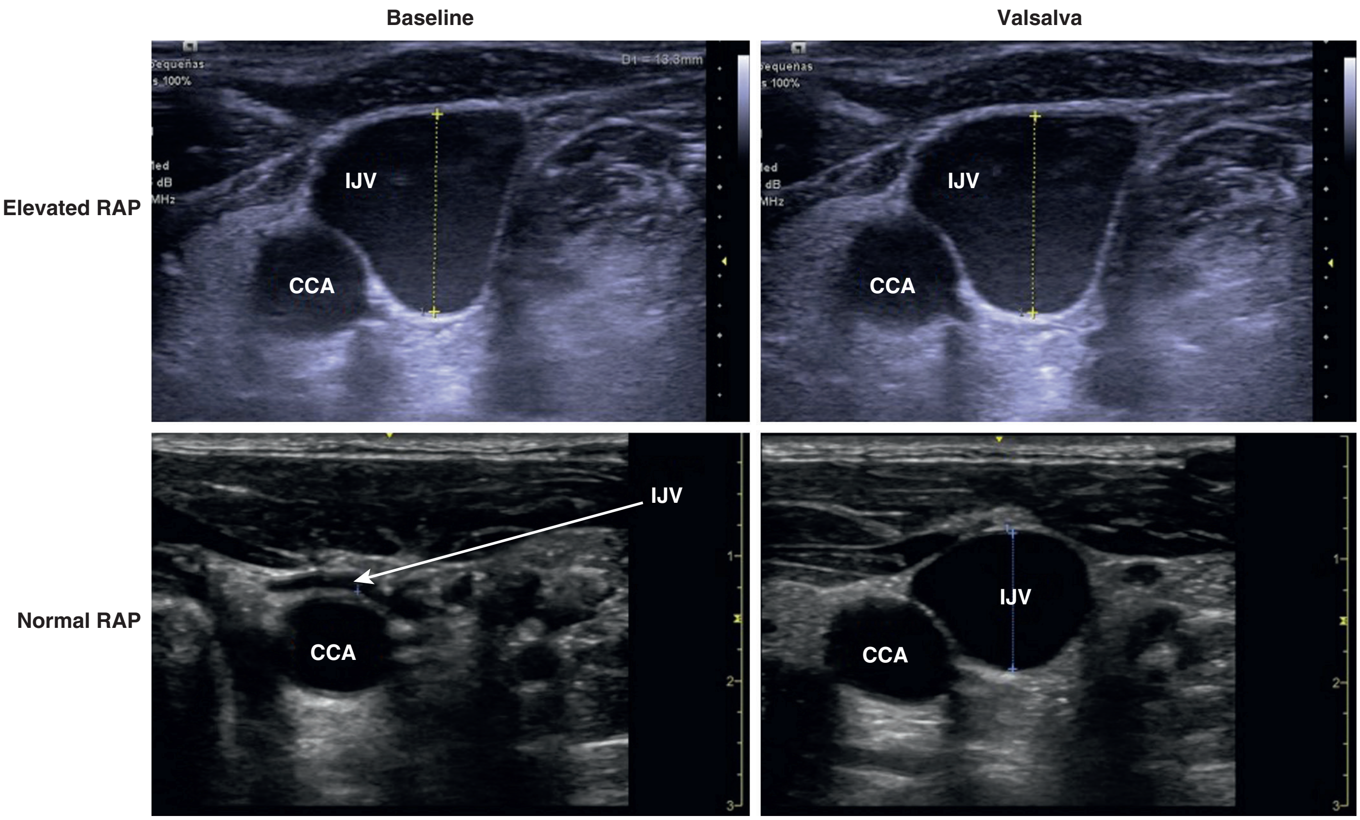


Fig. 26.19 Ultrasound images demonstrating internal jugular vein changes with Valsalva maneuver in a patient with elevated right atrial pressure and in a patient with normal right atrial pressure. *CCA,* Common carotid artery; *IJV,* internal jugular vein; *RAP,* right atrial pressure.

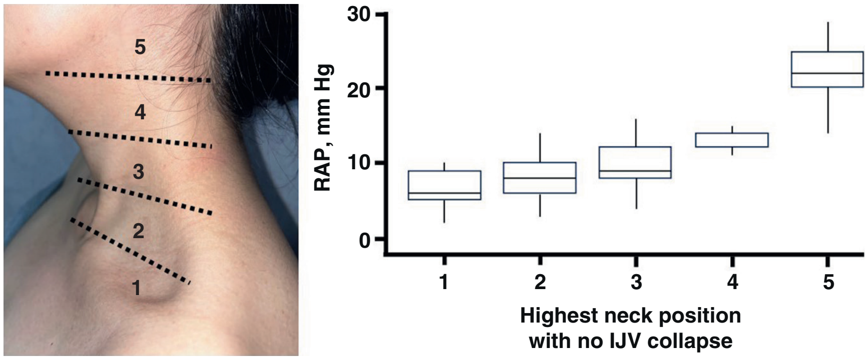


Fig. 26.20 Qualitative estimation of right atrial pressure (RAP) by localizing internal jugular vein (IJV) collapse point. The neck is divided into 5 visually estimated neck zones: 1, Clavicular level 2, lower third of neck; 3, middle third of neck; 4, upper third of neck; and 5, above the angle of the mandible. The highest point with no IJV collapse is recorded. This method shows good correlation with measured RAP, as shown in graph. (Adapted from Wang L, Harrison J, Dranow E, et al. Accuracy of ultrasound jugular venous pressure height in predicting central venous congestion. *Ann Intern Med.* 2022;175:344–351.)

Doppler interrogation of venous flow patterns in the peripheral organs has emerged as a promising tool for evaluating the degree of venous congestion.[49] Normal flow pattern in the central veins (e.g., IVC or hepatic veins) is pulsatile in nature. This "venous pulse" is a direct consequence of pressure changes in the right atrium that occur with every cardiac cycle.[50] In normal conditions, however, pulsatility is dampened by the highly compliant walls of the peripheral veins such that flow in distal venules and capillaries is no longer pulsatile (Fig. 26.21). Conditions that decrease venous compliance reduce their buffering capacity and augment the transmission of pulsatility to distal veins and even small venules.[51] For example, pulsatile saphenous venous flow is frequently observed in patients with advanced chronic peripheral venous insufficiency.[52] In patients with congestive heart failure, elevated RAP leads to increased venous volume and decreased venous compliance such that worsening degrees of venous congestion are associated with increased flow pulsatility in distal veins and venules.[53]

Intrarenal venous Doppler (IRVD) has been one of the most studied venous flow patterns in the assessment of venous congestion in heart failure. To perform this examination, pulsed wave Doppler is used to interrogate flow at the interlobar or arcuate renal veins at end-expiration. Given the proximity of interlobar arteries and veins, an arterial waveform is frequently obtained simultaneously when performing this examination.[51] Normal intrarenal venous flow (IRVF) is minimally pulsatile. With mild congestion, pulsatility increases (e.g., small amounts of volume administration can significantly increase intrarenal venous pulsatility in patients with heart failure even without noticeable changes in estimated central venous pressure.[54]) As venous congestion worsens, IRVF becomes highly pulsatile, even leading to flow reversal; however, this reversal is often difficult to discern due to its overlap with the concurrently acquired arterial waveform. Thus increasing degrees of venous congestion will manifest as increasingly interrupted IRVF (Fig. 26.22).[55] Pioneering work by Iida and colleagues[56] showed that interrupted IRVF reflects venous congestion in patients with heart failure and is strongly correlated with adverse clinical outcomes independent of RAP. These results were replicated in a large study involving patients with pulmonary hypertension where IRVF interruptions were associated with RAP, RV dysfunction, hydration status, and lower eGFR.[55] Interestingly, in this study, altered IRVF displayed a better correlation with intraabdominal pressure than with RAP, supporting the role of decreased venous compliance in pulsatility transmission.[55] While moderate increases in intraabdominal pressure are correlated with venous congestion in heart failure,[57] abdominal compartment syndrome can result in venous compression with IVC collapse and nonpulsatile venous flows.[58] Other studies involving patients with heart failure or undergoing cardiac surgery have consistently associated abnormal IRVD with AKI or adverse cardiovascular outcome.[59] Interrupted IRVF might be an early indicator of congestion in heart failure as it can be present without any noticeable changes in lung ultrasound or IVC in up to 34% of patients with interrupted flow.[60] However, IRVD can be time consuming and occasionally challenging to perform due to the small vessel size and respiratory motion, especially when using handheld ultrasound devices. Uninterpretable results are seen in up to 22% of captured images despite being performed by sonographers with dedicated training.[61] Also, alterations in kidney parenchyma such as advanced CKD, obstructive uropathy, or other structural abnormalities could alter the venous waveform.[53]

Given its larger size and favorable acoustic window, the main portal vein is an attractive alternative to IRVD for the evaluation of venous congestion.[62] Although it is physically closer to the RA than intrarenal veins, the portal circulation is relatively isolated from RAP transmission because of the liver sinusoid resistance.[63] Thus normal flow in the portal vein is nonpulsatile while it becomes increasingly pulsatile in the presence of worsening venous congestion[51] (Fig. 26.21). Increased portal vein pulsatility fraction has consistently been shown to be a marker of elevated RAP and to correlate with the development of congestive organ injury, such as congestive hepatopathy, enteropathy, encephalopathy, and AKI.[51] In a prospective study of patients undergoing cardiac surgery, increased portal vein pulsatility (>50%) after surgery was strongly associated with the subsequent development of AKI; the addition of portal vein pulsatility improved a baseline prediction model including perioperative risk factors and

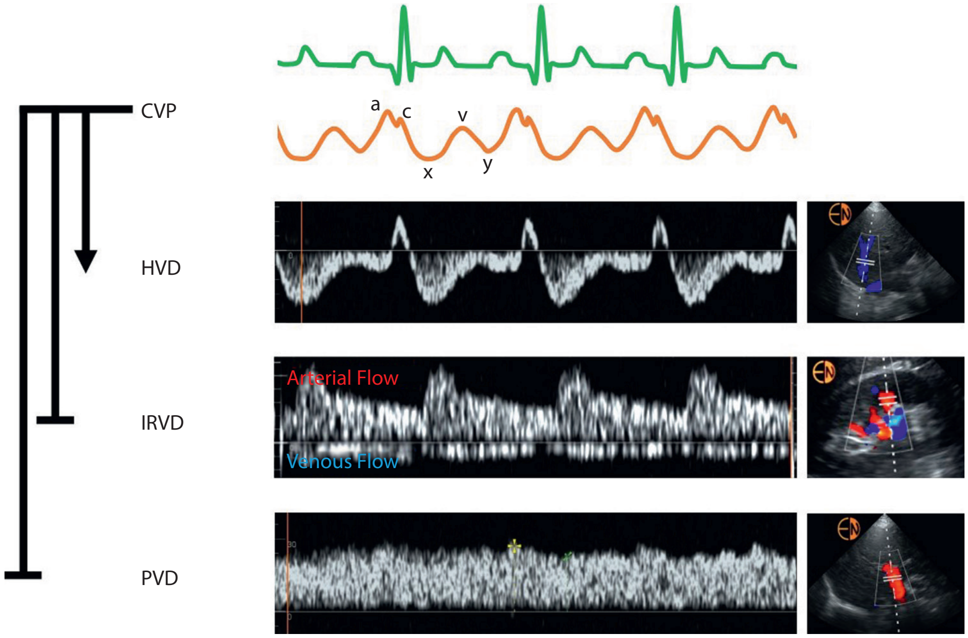


Fig. 26.21 Normal central venous pressure *(CVP)* waveform creates flow pulsatility in the central veins. Hepatic vein Doppler displays a pulsatile flow characterized by one retrograde wave (a wave) and two antegrade waves (s and d waves). Given normal venous compliance, this pulsatility is attenuated so that normal portal and intrarenal venous flow is only minimally pulsatile. *HVD,* hepatic vein Doppler; *IRVD,* intrarenal venous Doppler; *PVD,* portal vein Doppler.

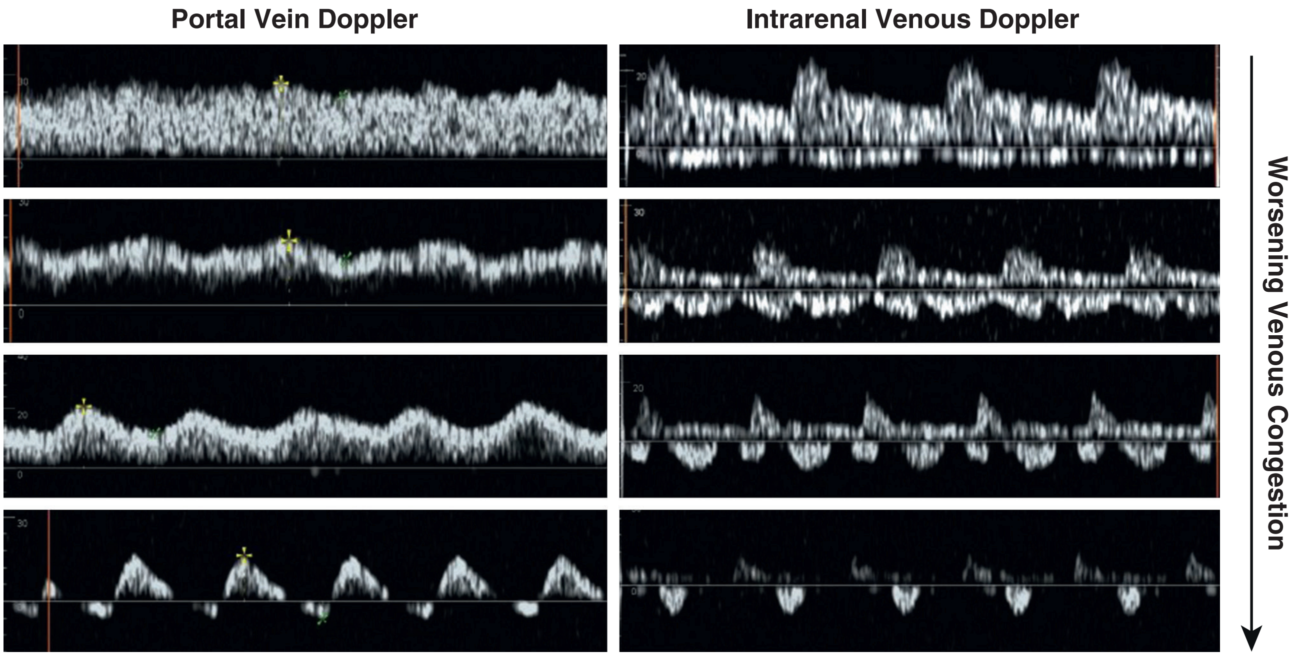


Fig. 26.22 Doppler waveforms of portal and intrarenal venous flow with worsening degrees of venous congestion. As congestion worsens, portal vein pulsatility increases from normal (>30% pulsatility) to severely abnormal (>50%–70% pulsatility). Intrarenal venous flow becomes progressively more interrupted with worsening congestion.

measured central venous pressure (CVP).[64] In patients with underlying pulmonary hypertension, portal vein pulsatility was shown to have excellent correlation and agreement with IRVF interruption time, and it was easier to obtain and showed high reproducibility even among nonexpert sonographers.[62] However, there are some important caveats to consider when performing this examination. Patients with advanced cirrhosis can present with dampened pulsatility despite severe congestion or increased pulsatility due to arterioportal shunts in the absence of heart failure.[53] Portal vein flow is frequently pulsatile in young healthy subjects with no venous congestion likely due to increased vagal tone and reduced sinusoidal resistance,[51] as this is not frequently seen in patients with cardiovascular disease.[62] Given the known caveats of IRVD and portal vein Doppler, congestion assessment accuracy can be increased by acquiring multiple data points alongside other indicators of RAP such as IJV or IVC ultrasound and FoCUS.[65]

Several studies have used venous Doppler as a monitoring tool during decongestive treatment. In hospitalized patients with heart failure, improvement in IRVF pattern can be observed in up to 60% of patients and is associated with a significantly better prognosis.[61,66] Improvement in portal vein pulsatility has also been observed following diuretic treatment in heart failure patients.[67] In patients with type 1 cardiorenal syndrome, portal vein alterations were completely resolved with diuretic treatment while there was no change in IVC evaluation in those with right heart failure.[68] A study involving patients in a cardiac ICU showed that altered portal and IRVD waveforms were good predictors of adequate response to diuretic therapy assessed by a clinical congestion score. Other echocardiographic parameters including IVC and hepatic vein Doppler were not useful.[69] While venous congestion is a function of increased RAP, it is possible that monitoring changes in venous flow patterns could provide more actionable information than monitoring pressure. Changes in venous volume are more sensitive indicators of volume removal than changes in measured CVP.[70] An average decrease in RAP of only 4 mm Hg was observed in the ESCAPE trial after 48 hours of diuretic treatment;[71] this change would be hard to detect by IVC or IJV ultrasound.[46] Conversely, average portal vein pulsatility fraction decreased more than 50% after 48 to 72 hours of diuretic treatment in patients with severe congestion.[68] Finally, normal venous Doppler patterns can be observed in patients with pulmonary hypertension even with considerable elevations in RAP presumably due to chronic adaptations resulting in increased venous compliance.[55] While the RAP acts as a back-pressure, highly pulsatile Doppler patterns in the setting of decreased splanchnic venous compliance could constitute an additional pulsatile load for the target organs.[72] Interventional trials that specifically focus on addressing venous Doppler alterations in various heart failure populations are awaited.

It is fundamental to understand that venous Doppler is not a tool aimed at determining *volume status*. Because venous congestion is a function of increased RAP and venous compliance, patients without cardiac dysfunction may not display altered venous Doppler patterns even when severely volume overloaded.[73] Despite the high prevalence of volume overload, the prevalence of venous congestion assessed by venous Doppler in nonselected critically ill patients is low and is not predictive of AKI;[74] this finding is expected because there are more frequent causes of renal deterioration besides congestion in this population. On the other hand, even severe venous congestion can be observed in the absence of volume overload in conditions like cardiac tamponade, large pulmonary embolism, or tension pneumothorax.[73] It is crucial to incorporate Doppler findings into the assessment along with the patient's clinical presentation and underlying cardiovascular status.

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